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Daytime

$$1.) i_2 = \frac{\frac{32}{50} \cdot \frac{1}{2} + \frac{18-60}{50} = \frac{108-V_0}{50}$$

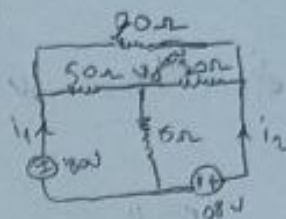
$$i_2 + \frac{4}{5} = \frac{8}{5} \quad i_2 = \frac{4}{5} \text{ A}$$

$$i_1 = \frac{180-108}{50} + \frac{180-V_0}{50}$$

$$i_1 = \frac{4}{5} + \frac{12}{5} = \frac{16}{5} \text{ A}$$

$$\boxed{\begin{matrix} i_1 = \frac{16}{5} \text{ A} \\ i_2 = \frac{4}{5} \text{ A} \end{matrix}}$$

$$V_0 = 60 \text{ V}$$



Find i_1 and i_2 .

$$\frac{V_0}{15} = \frac{180-V_0}{50} + \frac{108-V_0}{50}$$

$$10V_0 = 540 - 3V_0 + 540 - 5V_0$$

$$18V_0 = 1080$$

$$V_0 = 60$$

2.) Constraint equation $i_x = \frac{V_2 - 0}{25}$

KCL for node 1

$$\frac{V_1 - 20}{4} + \frac{V_1 - V_2}{1} + 40i_x = 0 \quad 25V_1 + 12V_2 = 100$$

KCL for node 2

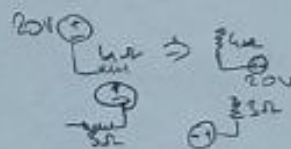
$$\frac{V_2 - V_1}{1} + \frac{V_2}{25} + \frac{V_2 - V_3}{2} = 0 \quad -50V_1 + 77V_2 - 25V_3 = 0$$

KCL for node 3

$$\frac{V_3 - V_2}{2} + \frac{V_3 - 80}{3} - 40i_x = 0 \quad -63V_2 + 25V_3 = 800$$



Write the node-voltage equations and the constraint equation.



3.) Constraint equation $i_y = i_a + i_b \rightarrow$ special case of mesh-current method, $i_c = 4i_a$

KVL for loop 1

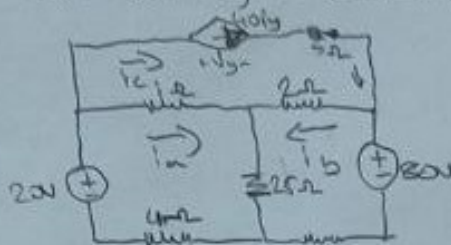
$$-20 + (i_a - i_c) + 25(i_a + i_b) + 4i_a = 0 \quad 30i_a + 25i_b - i_c = 20$$

KVL for loop 2

$$2(i_b + i_c) + 25(i_a + i_b) + 3i_b - 80 = 0 \quad 25i_a + 30i_b + 2i_c = 80$$

KVL for loop 3

$$(i_c - i_a) + V_y + 5i_c + 2(i_c + i_b) = 0$$



Write the mesh-current equations and the constraint equation.